

Amir Kooshan Fattah Hesari

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Professional Summary

Junior AI Engineer focused on building useful machine-learning systems, not only training models. Experienced with **Python**, **PyTorch**, **LLM tools**, and data-driven experimentation.

Education

Sharif University of Technology [🔗](#)

Sept 2022– 2026(expected)

B.Sc. in Electrical Engineering - Major GPA 17.1/20

Alborz High School

2019– 2022

Diploma in Mathematics and Physics

- Affiliated with National Organization for Development of Exceptional Talent (**NODET** [🔗](#))

Related Courses

- | | | | |
|-------------------------------|---------|-------------------------------|-----------|
| ◦ Deep Reinforcement Learning | (19/20) | ◦ High Dimensional Statistics | (18.5/20) |
| ◦ Deep Generative Models | (20/20) | ◦ Convex Optimization | (17.9/20) |

Experience

research at CUHK *research, supervised*

Onsite

by **Dr. Qiuqiang Kong** [🔗](#)

July. 2025 - Sep. 2025

- Used **Pre-trained Audio Neural Networks (PANNs)** and transfer learning to study anomaly detection in **EEG signals** for neurological monitoring.
- Built a cleaner **data pipeline** for collection, annotation, and preprocessing. The workflow supported large-scale dataset preparation for AI experiments and reduced manual checking.

MLL Lab *research, supervised*

Onsite

by **Dr. Soleymani Baghshah** [🔗](#)

Jan. 2026 -

- Worked on **Meta Reinforcement Learning** methods for improving task generalisation and fast adaptation to unseen environments
- Used neural memory blocks inspired by **test-time training**, enabling the agent to update its internal representation during interaction rather than relying only on fixed learned parameters
- Incorporated ideas from **continual learning** to help the agent retain useful knowledge across tasks while adapting to new and changing environments
- Studied adaptation under **non-stationary dynamics**, where the agent must use limited recent experience to infer task changes and adjust its behaviour accordingly

Projects

Multi-Armed Bandits for Online Decision-Making *Python, NumPy, Pandas, Matplotlib, Git*

- Implemented bandit algorithms (ϵ -Greedy, UCB, Thompson Sampling) for exploration-exploitation optimization
- Evaluated performance under non-stationary settings and applied to financial trading strategies

Deep Generative Models for Adversarial Attacks on LLMs *Python, PyTorch, Transformers, Deep Generative Models*

- Used deep generative models to synthesize adversarial prompts that probe robustness and failure modes of large language models
- Evaluated attack success rates under different safety filters and investigated mitigation strategies to improve

robustness

Seizure Prediction Using Machine Learning Methods *Python, Matlab, Machine Learning, Signal Processing*

- Developed seizure prediction using spectral entropy features from EEG signals
- Applied KNN, SVM, regression, and ANNs for neurological monitoring applications

Conjugate Gradient Methods for Hessian-Free Optimization *Python, NumPy, Matplotlib, CVXPY*

- Implemented convex optimization algorithms from scratch achieving $\leq 10^{-6}$ optimality gaps
- Applied to portfolio optimization and sparse regression with $3\text{-}5\times$ speedups over standard solvers

Honors and Awards

- **Nationwide University Entrance Exam (Konkur) - 2022** ranked 155 out of 120,000+ participants

Skills

Programming: Python for machine learning, data analysis, and research code. **NumPy, Pandas, SciPy, Scikit-learn, Matplotlib**

Machine Learning & Deep Learning: PyTorch , Jax, neural-network training, model evaluation, feature engineering, and experiment design. **reinforcement learning, deep generative models**, and EEG/signal-processing applications.

LLM & AI Tools: Hugging Face Transformers, LangChain, prompt engineering, embeddings, and basic RAG workflows.

Backend & Software Development: Django, Java, OOP, C/C++

Language

- English : Fluent
- Persian : Native
- German : Basic